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Inventor(s)	Hashimoto; Kohei et al.

Protector and wire harness

Abstract

A protector includes a shell portion that is formed from a resin and covers an outer side of an electric wire and a protection portion that dampens an impact applied from outside the protector. The protection portion is formed from a resin and formed integrally with the shell portion.

Inventors:	Hashimoto; Kohei (Yokkaichi, JP), Inoue; Hiroshi (Yokkaichi, JP), Suga; Takao (Yokkaichi, JP), Ito; Takahiro (Yokkaichi, JP), Takahashi; Hayate (Yokkaichi, JP), Takahashi; Makoto (Toyoake, JP), Satoh; Kengo (Toyota, JP), Shimamura; Takuma (Nisshin, JP), Uchioka; Yusuke (Okazaki, JP)
Applicant:	SUMITOMO WIRING SYSTEMS, LTD. (Mie-ken, JP); TOYOTA JIDOSHA KABUSHIKI KAISHA (Aichi-ken, JP)
Family ID:	77613480
Assignee:	SUMITOMO WIRING SYSTEMS, LTD. (Mie-ken, JP); TOYOTA JIDOSHA KABUSHIKI KAISHA (Aichi-ken, JP)
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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a protector and a wire harness.

BACKGROUND ART

(2) A typical wire harness arranged in a vehicle may include an electric wire and a shell member that covers the electric wire. Patent Literature 1 describes an example of such type of a wire harness that further includes a protection member to protect the electric wire from an impact applied from outside the wire harness. In this case, the shell member and the protection member form a protector that protects the electric wire.

(3) In the wire harness described in Patent Literature 1, the electric wire is accommodated in a groove of the protection member. Further, the wire, together with the protection member, is accommodated in a cover serving as a shell member that is disposed in part of an interior member of the vehicle.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Laid-Open Patent Publication No. 2002-142326

SUMMARY OF INVENTION

Technical Problem

(5) The wire harness of Patent Literature 1 is assembled by attaching the electric wire to the protection member and then covering the protection member and the electric wire with the cover. To improve assembling workability, there is a demand for a reduction in the number of parts.

(6) An objective of the present disclosure is to provide a protector and a wire harness that reduce the number of parts.

Solution to Problem

(7) A protector according to the present disclosure includes a shell portion and a protection portion. The shell portion is formed from a resin and covers an outer side of an electric wire. The protection portion dampens an impact applied from outside the protector. The protection portion is formed from a resin and formed integrally with the shell portion.

(8) A wire harness according to the present disclosure includes an electric wire and the above-described protector. The electric wire is electrically connected to an electric power supply that drives a power source for moving a vehicle. The protector includes the shell portion that covers the outer side of the electric wire.

Advantageous Effects of Invention

(9) The protector and the wire harness according to the present disclosure reduces the number of parts.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram showing a wire harness in accordance with an embodiment laid out in a vehicle.

(2) FIG. 2 is a schematic diagram showing the wire harness in accordance with the embodiment laid out in the vehicle.

(3) FIG. 3 is a perspective view of the wire harness in accordance with the embodiment.

- (4) FIG. 4 is a perspective view of a protector in accordance with the embodiment.
- (5) FIG. 5 is a front view of the protector in accordance with the embodiment.
- (6) FIG. 6 is a cross-sectional view of the protector in accordance with the embodiment.
- (7) FIG. 7 is a cross-sectional view of the protector in accordance with the embodiment.
- (8) FIG. 8 is a cross-sectional view of the protector in accordance with the embodiment.
- (9) FIG. 9 is an enlarged view showing part of the protector in accordance with the embodiment.

DESCRIPTION OF EMBODIMENT

Description of Embodiment of Present Disclosure

- (10) An embodiment of the present disclosure will now be described.
- (11) A protector of the present disclosure is as follows.
- (12) (1) A protector including a shell portion and a protection portion. The shell portion is formed from a resin and covers an outer side of an electric wire. The protection portion dampens the impact applied from outside the protector. The protection portion is formed from a resin and formed integrally with the shell portion.
- (13) With the above structure, the protection portion dampens the impact applied from outside of the protection portion. This dampens the impact received by the electric wire. Further, the shell portion is formed integrally with the protection portion. Thus, the protector has fewer parts than a protector in which a shell portion and a protection portion are separate parts.
- (14) (2) Preferably, the protection portion includes cells extending parallel to one another.
- (15) With the above structure, the electric wire receives the impact dampened by the protection portion, which includes the parallel cells.
- (16) (3) Preferably, the protection portion disperses and absorbs the impact applied to the protection portion.
- (17) With the above structure, the protection portion disperses and absorbs the impact from outside the protector to dampen the impact received by the electric wire.
- (18) (4) Preferably, the shell portion is gutter-shaped and includes two opposing side walls and a bottom wall connecting an end of each of the two side walls. Preferably, the protection portion is integrated with an outer surface of the bottom wall. Preferably, the cells extend in an opposing direction of the two side walls.
- (19) The above structure limits the length of the cells. This facilitates manufacture of the protector with a mold.
- (20) (5) Preferably, the protection portion has a honeycomb structure in which the cells are hexagonal.
- (21) With the above structure, the protection portion further effectively dampens an impact.
- (22) (6) Preferably, a part of the bottom wall that is integrated with the protection portion is curved in an arcuate manner and bulged toward the protection portion in a cross section that is perpendicular to a direction in which the shell portion extends.
- (23) The above structure allows the protector to withstand a larger impact applied to the protection portion than when the part of the bottom wall integrated with the protection portion extends straight in a cross section that is perpendicular to the direction in which the shell portion extends.
- (24) (7) Preferably, the protection portion is integrated with a part of an outer surface of the shell portion. Preferably, the protection portion includes a flat side surface facing a side opposite to the shell portion and exposed to the outside.
- (25) With the above structure, when an object strikes the side surface of the protection portion, which is located at the side opposite to the shell portion, the force of the object will be applied to the protection portion in a direction perpendicular to the side surface.
- (26) A wire harness of the present disclosure is as follows.
- (27) (8) A wire harness including an electric wire and the above-described protector. The electric wire is electrically connected to an electric power supply that drives a power source for moving a vehicle. The protector includes the shell portion that covers the outer side of the electric wire.

(28) With the above structure, the electric wire is connected to the electric power supply that drives a power source for moving the vehicle. Thus, a high voltage may be applied to the electric wire. The protector dampens the impact received by such an electric wire.

Detailed Description of Embodiment of Present Disclosure

(29) A specific example of the protector and the wire harness in accordance with the present disclosure will now be described with reference to the drawings. The present invention is not limited to the illustrated examples and is intended to be defined by the claims and their equivalents, and all variations within the scope of the claims and their equivalents. In the appended drawings, elements may not be drawn to scale.

(30) An embodiment of a protector and a wire harness will now be described.

(31) As shown in FIGS. 1 and 2, a wire harness 20 electrically connects electric components 11 installed in a vehicle 10, such as a hybrid car or an electric vehicle. FIG. 1 shows only two of the electric components 11 arranged in the vehicle 10. FIG. 2 is a schematic diagram of the vehicle 10 as viewed from the front side. For example, at the front part of the vehicle 10, the wire harness 20 is laid out to extend from the right side to the left side of the vehicle 10. Further, the wire harness 20 extends under the floor of the vehicle 10 to the rear part of the vehicle 10.

(32) As shown in FIG. 3, the wire harness 20 includes electric wires 21 and a protector 22 that covers the outer side of the electric wires 21. In the present embodiment, the wire harness 20 includes five electric wires 21. Each electric wire 21 electrically connects two electric components 11. For example, as shown in FIGS. 3 and 1, one of the electric wires 21 electrically connects an inverter 12 and a battery 13. The inverter is located at the front part of the vehicle 10. The battery 13 is located toward the rear of the vehicle 10 from the inverter 12. The inverter 12 is connected to a wheel driving motor (not shown) serving as a power source for moving the vehicle. In other words, this single electric wire 21 is electrically connected to the battery 13 that is an electric power supply for driving the motor corresponding to a power source for moving the vehicle. The inverter 12 generates alternating current from direct current of the battery 13 and supplies the generated alternating current to the motor. The battery 13 is a battery capable of supplying, for example, a hundred to several hundreds of volts.

Structure of Protector 22

(33) As shown in FIG. 3, the protector 22 includes a shell portion 31 and a protection portion 41. The shell portion 31 covers the outer side of the electric wires 21. The protection portion 41 dampens the impact applied from outside the protector 22. The protector 22 is formed from resin. The resin material of the protector 22 is, for example, polyamide or the like. The protection portion 41 is formed integrally with the shell portion 31. A mold for forming the protector 22 is filled with resin. Solidification of the resin forms the protector 22 with the protection portion 41 integrated with the shell portion 31.

Structure of Shell Portion 31

(34) As shown in FIG. 4, the shell portion 31 is gutter-shaped and includes two opposing side walls 32 and 33 and a bottom wall 34. The bottom wall 34 connects an end of each of the two side walls 32 and 33. In the drawing, double-headed arrow X1 indicates the direction in which the shell portion 31 extends.

(35) The side walls 32 and 33 are flat. The side walls 32 and 33 are spaced apart from each other in a thickness direction of the side walls 32 and 33 and oppose each other in the thickness direction. That is, in the protector 22 of the present embodiment, an opposing direction X2 of the two side walls 32 and 33 coincides with the thickness direction of the side walls 32 and 33. The opposing direction X2 may differ between different parts of the shell portion 31 in accordance with the direction in which the shell portion 31 extends.

(36) The bottom wall 34 is formed integrally with the side walls 32 and 33. The bottom wall 34 is belt-shaped. The side walls 32 and 33 respectively project from two widthwise ends of the bottom wall 34 in a thickness direction of the bottom wall 34. In the present embodiment, the shell portion

31 is bracket-shaped in a cross section extending in the opposing direction **X2**, or a cross section that is perpendicular to the direction in which the shell portion **31** extends. The side walls **32** and **33** and the bottom wall **34** define a groove-like electric wire accommodation portion **35** where the electric wires **21** are disposed in the shell portion **31**.

(37) As shown in FIG. 3, the electric wires **21** are accommodated in the electric wire accommodation portion **35** so that the shell portion **31** covers the outer side of the electric wires **21**. The electric wires **21** are arranged in the electric wire accommodation portion **35** such that the longitudinal direction of the electric wires **21** is perpendicular to the opposing direction **X2** and such that the electric wires **21** extend in the direction in which the shell portion **31** extends. A lid **51** is attached to the end of each of the side walls **32** and **33** that is opposite to the end connected to the bottom wall **34**. The lid **51** covers the opening of the electric wire accommodation portion **35**. The lid **51** is formed from resin. The lid **51** restricts separation of the electric wires **21** from the shell portion **31**.

Structure of Protection Portion **41**

(38) As shown in FIG. 5, the protection portion **41** is integrated with an outer surface of the shell portion **31**. The protection portion **41** of the present embodiment disperses and absorbs the impact applied by an object striking the protection portion **41**.

(39) As shown in FIGS. 5 and 8, the protection portion **41** is formed integrally with an outer surface of the bottom wall **34**. FIG. 8 is a cross-sectional view of the protector **22** taken at a central part of the bottom wall **34** in the widthwise direction along a plane that is perpendicular to the opposing direction **X2**. In FIGS. 5 and 8, the opposing direction **X2** where the protection portion **41** is located in the shell portion **31** is perpendicular to the plane of the drawing. The bottom wall **34** includes a part that is recessed toward the inner side of the shell portion **31** as viewed in the opposing direction **X2**. Accordingly, the outer surface of the bottom wall **34** defines a protection recess **36**. The protection portion **41** is arranged in the protection recess **36** and formed integrally with peripheral surfaces in the protection recess **36**. Specifically, the protection portion **41** is formed integrally with a bottom **36a** of the protection recess **36** that is part of the bottom wall **34**. Further, the two ends of the protection portion **41** in the direction in which the shell portion **31** extends are formed integrally with two side walls **36b** of the protection recess **36** in the direction in which the shell portion **31** extends.

(40) As shown in FIGS. 6 and 7, the bottom **36a** is a part of the bottom wall **34** that is integrated with the protection portion **41**. In a cross section that is perpendicular to the direction in which the shell portion **31** extends, the bottom **36a** is curved in an arcuate manner and bulged toward the protection portion **41**. In other words, in a cross section of the protector **22** that is perpendicular to the direction in which the shell portion **31** extends, the bottom **36a** is arched and bulged toward the outer side of the shell portion **31**.

(41) The protection portion **41** includes an impact receiving wall **42** at an end in a direction perpendicular to the opposing direction **X2** in the protection portion **41** and the direction in which the shell portion **31** extends. The impact receiving wall **42** is disposed at the side of the protection portion **41** opposite to the shell portion **31**. The impact receiving wall **42** is flat. Further, the impact receiving wall **42** extends in a direction parallel to the direction in which the shell portion **31** extends. The outer surface of the impact receiving wall **42** defines an impact receiving surface **43**. The impact receiving surface **43** faces a side opposite to the shell portion **31** and is exposed to the outside of the protector **22**. The impact receiving surface **43** extends over a range of the protector **22** that expected to be vulnerable to a striking object. In the present embodiment, the impact receiving surface **43** faces the side opposite to the bottom **36a** of the protection recess **36** in the protection portion **41**. Further, the impact receiving surface **43** is exposed to the outside of the protector **22** over the open end of the protection recess **36** in a direction perpendicular to the opposing direction **X2**. The impact receiving surface **43** is flat. In the present embodiment, as viewed in the opposing direction **X2**, the impact receiving surface **43** extends in the opposing

direction X2 and is parallel to the bottom 36a of the protection recess 36. Further, as viewed in the opposing direction X2, the impact receiving surface 43 extends parallel to the part of the shell portion 31 that is integrated with the protection portion 41. The protector 22 is arranged in the vehicle 10 with the impact receiving wall 42 facing the direction from which an object is expected to strike the protector 22.

(42) As shown in FIGS. 5 to 8, the protection portion 41 includes the cells 44 extending parallel to one another. FIG. 6 is a cross-sectional view of the protector 22 taken along line 6-6 shown in FIG. 5 that is perpendicular to the direction in which the shell portion 31 extends. FIG. 7 is a cross-sectional view of the protector 22 taken along line 7-7 shown in FIG. 5 that is perpendicular to the direction in which the shell portion 31 extends. The cells 44 extend in the opposing direction X2 in the part where the protection portion 41 is formed integrally with the shell portion 31. Further, the cells 44 are arranged at regular intervals.

(43) As shown in FIG. 9, as viewed in the opposing direction X2, the cells 44 have the form of a regular hexagon. In FIG. 9, the opposing direction X2 is perpendicular to the plane of the drawing. Thus, the protection portion 41 has a honeycomb structure in which the cells 44 have the form of a regular hexagon. As viewed in the opposing direction X2, five rows of the cells 44, with the cells 44 of each row arranged next to one another in the direction in which the shell portion 31 extends, are formed between the impact receiving surface 43 and the bottom 36a of the protection recess 36. In FIG. 9, the cells 44 of each row are arranged adjacent to one another in a sideward direction that is parallel to the planar direction of the impact receiving surface 43. As viewed in the opposing direction X2, the cells 44 located at ends of the protection portion 41 may not have the shape of a complete regular hexagon and may have the shape of part of a regular hexagon. Further, the cells 44 located at the ends of the protection portion 41 may include ones extending in the opposing direction X2 that are not through holes depending on the forms of the bottom 36a and the side walls 36b of the protection recess 36.

(44) The protection portion 41 includes first ribs 45 and second ribs 46 extending between adjacent cells 44. The first ribs 45 and the second ribs 46 are post-shaped and extend in the opposing direction X2. As viewed in the opposing direction X2, the first ribs 45 extend perpendicular to the impact receiving surface 43. Further, in the present embodiment, the first ribs 45 extend perpendicular to the bottom 36a of the protection recess 36 as viewed in the opposing direction X2. Two second ribs 46 branch off from each first rib 45, and each second rib 46 is tilted with respect to the first rib 45 as viewed in the opposing direction X2. Each cell 44 is defined by two first ribs 45 and four second ribs 46. Among the five rows of the cells 44, the first ribs 45 of the cells 44 in the first row from the impact receiving wall 42 are connected to and integrated with the impact receiving wall 42. The first ribs 45 and the second ribs 46 support the impact receiving wall 42 where an object is expected to strike.

(45) In the present embodiment, the rib thickness T1 of the first rib 45 and the rib thickness T2 of the second rib 46 are set to be the same value. Further, the pitch P between adjacent cells 44 is set to be a greater value than the rib thickness T1, T2. A change in the rib thickness T1, T2 and the pitch P will result in a change in the cross-sectional area of the portion that supports the impact receiving wall 42, which receives the impact of a striking object. The cross-sectional area of the portion that supports the impact receiving wall 42 corresponds to the cross-sectional area of the protection portion 41 taken along a plane that is parallel to the impact receiving surface 43.

(46) Preferably, the rib thickness T1, T2 is 1.5 mm or greater and 3.0 mm or less. Further preferably, the rib thickness T1, T2 is 1.5 mm or greater and 2.0 mm or less. Moreover, it is preferred that the pitch P be greater than the rib thickness T1, T2 and greater than or equal to 5 mm. Further preferably, the pitch P is greater than 7.0 mm and less than 9 mm.

(47) Simulations were conducted with a protector including a protection portion of which the structure is the same as the protection portion 41 of the present embodiment. In the simulations, an impact having an expected force was applied from outside to the protectors having various rib

thickness **T1**, **T2** and pitch **P**. As a result, preferred combinations of the rib thickness **T1**, **T2** and the pitch **P** were obtained as follows. For example, when the rib thickness **T1**, **T2** is 1.5 mm, a preferred pitch **P** is 7.25 mm. When the rib thickness **T1**, **T2** is 2.0 mm, a preferred pitch **P** is 7.25 mm. When the rib thickness **T1**, **T2** is 2.5 mm, a preferred pitch **P** is 8.0 mm. When the rib thickness **T1**, **T2** is 3.0 mm, a preferred pitch **P** is 8.75 mm.

Operation

(48) The operation of the present embodiment will now be described.

(49) When an object strikes the impact receiving surface **43**, the impact is applied to the impact receiving surface **43** by the object. The impact receiving wall **42** disperses the impact with the first ribs **45** that are integrated with the impact receiving wall **42**. Then, the impact dispersed with the first ribs **45** is transmitted through the first ribs **45** and the second ribs **46** in the protection portion **41**, thereby dampening the impact. In this manner, the protection portion **41** disperses and absorbs the energy of the impact applied from the outside of the protector **22** to the impact receiving surface **43**.

(50) The present embodiment has the following advantages.

(51) (1) The protector **22** includes the shell portion **31** and the protection portion **41**. The shell portion **31** is formed from resin and covers the outer side of the electric wires **21**. The protection portion **41** dampens the impact applied from outside the protector **22**. The protection portion **41** is formed from resin and formed integrally with the shell portion **31**.

(52) With the above structure, the protection portion **41** dampens the impact applied to the protection portion **41** by an object striking the protection portion **41**. This dampens the impact received by the electric wires **21**. Further, the shell portion **31** is formed integrally with the protection portion **41**. Thus, the protector **22** has fewer parts than a protector in which a shell portion and a protection portion are separate parts. This improves assembling workability of the wire harness **20**. Also, the manufacturing costs are reduced as compared with when a protection material is wrapped around the electric wires and disposed in a shell portion with the electric wires to protect the electric wires from an impact.

(53) (2) The protection portion **41** includes the cells **44** extending parallel to one another. Accordingly, the electric wires **21** receive the impact dampened by the protection portion **41** including the cells **44** that extend parallel to one another. Thus, the protection portion **41** that has a simple structure dampens the impact received by the electric wires **21**.

(54) (3) The protection portion **41** disperses and absorbs the impact applied to the protection portion **41**. The protection portion **41** disperses and absorbs the impact applied from outside the protector **22** and dampens the impact received by the electric wires **21**. The protection portion **41** dampens the impact on the electric wires **21** without increasing the protection portion in size or using an expensive resin material in the protection portion.

(55) (4) The shell portion **31** is gutter-shaped and includes the two opposing side walls **32** and **33** and the bottom wall **34**. The bottom wall **34** connects the end of each of the two side walls **32** and **33**. The protection portion **41** is integrated with the outer surface of the bottom wall **34**. The cells **44** extend in the opposing direction **X2** of the two side walls **32** and **33**.

(56) The above structure limits the length of the cells **44** as compared with when the cells extend in the direction in which the shell portion **31** extends. This facilitates manufacture of the protector **22** with a mold.

(57) (5) The protection portion **41** has a honeycomb structure in which the cells **44** are hexagonal. Thus, the protection portion **41** further effectively dampens an impact.

(58) (6) The part of the bottom wall **34** that is integrated with the protection portion **41** is curved in an arcuate manner and bulged toward the protection portion **41** in a cross section that is perpendicular to the direction in which the shell portion **31** extends. In the present embodiment, the bottom **36a** of the protection recess **36** is arched and bulged toward the outer side of the shell portion **31** in a cross section that is perpendicular to the direction in which the shell portion **31**

extends. This allows the protector **22** to withstand a larger impact applied from the outside of the protector **22** to the protection portion **41** than when the part of the bottom wall **34** integrated with the protection portion **41** extends straight in a cross section that is perpendicular to the direction in which the shell portion extends.

(59) (7) The protection portion **41** is integrated with a part of the outer surface of the shell portion **31**. The protection portion **41** includes a flat side surface facing a side opposite to the shell portion **31** and exposed to the outside of the protector **22**. Accordingly, when an object strikes the side surface of the protection portion **41**, which is located at the side opposite to the shell portion **31**, the force of the object will be applied to the protection portion **41** in a direction perpendicular to the side surface.

(60) In the present embodiment, the impact receiving surface **43** corresponds to the side surface of the protection portion **41** that faces the side opposite to the shell portion **31** and is exposed to the outside of the protector **22**. As viewed in the opposing direction X2, the impact receiving surface **43** is flat and extends parallel to the bottom **36a** in the direction in which the shell portion **31** extends. Further, as viewed in the opposing direction X2, the first ribs **45** are perpendicular to the impact receiving surface **43**. Therefore, when an object strikes the impact receiving surface **43**, the force of the object will be applied to the protection portion **41** in a direction parallel to the first rib **45** as viewed in the opposing direction X2. In this manner, the protection portion **41** effectively disperses and dampens the impact received by the electric wires **21**.

(61) (8) The wire harness **20** includes the electric wire **21** and the protector **22**. The electric wire **21** is connected to the battery **13** that drives the power source for moving the vehicle. The protector **22** includes the shell portion **31** that covers the outer side of the electric wire **21**. With this structure, the electric wire **21** is connected to the battery **13** that drives the power source for moving the vehicle. Thus, a high voltage may be applied to the electric wire **21**. In the wire harness **20**, the protector **22** dampens the impact received by such an electric wire **21**.

(62) (9) When the rib thickness T1 of the first ribs **45** and the rib thickness T2 of the second rib **46** are set to 1.5 mm or greater and 2.0 mm or less, the protector **22** is easily formed with a mold. This facilitates mass-production of the protector **22**.

(63) (10) The protector **22** of the present embodiment has a structure in which the protection portion **41**, which dampens an impact applied from outside the protector **22**, is formed integrally with the shell portion **31**, which covers the outer side of the electric wires **21**. Accordingly, depending on the resin material for the protector **22**, the structure of the present embodiment may reduce costs when compared to a conventional structure in which a protection member, which protects the electric wire from an impact applied from outside the protection member, is formed separately from a shell member, which covers the electric wire.

(64) The present embodiment may be modified as follows. The present embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

(65) In the above embodiment, as viewed in the opposing direction X2, the protection portion **41** includes five rows of the cells **44** between the impact receiving surface **43** and the bottom **36a** of the protection recess **36**. The cells **44** are arranged adjacent to one another in the direction in which the shell portion **31** extends. However, the number of the rows of the cells **44** does not have to be five and may be four or less or six or greater as viewed in the opposing direction X2. Preferably, there is more than one row of the cells **44** arranged next to one another in the direction in which the shell portion **31** extends between the impact receiving surface **43** and the bottom **36a** of the protection recess **36** as viewed in the opposing direction X2. In this way, the impact applied from outside to the protection portion **41** is absorbed more effectively by the protection portion **41** as compared with when only one row is formed. This further dampens the impact received by the electric wires **21**.

(66) In the above embodiment, the impact receiving surface **43**, that is, the side surface of the

protection portion **41**, which faces the side opposite to the shell portion **31** and is exposed to the outside, is flat. However, the impact receiving surface **43** does not necessarily have to be a flat and may be, for example, curved or stepped.

(67) In the above embodiment, the bottom **36a** of the protection recess **36** is arched and bulged toward the outer side of the shell portion **31** in a cross section that is perpendicular to the direction in which the shell portion **31** extends. However, the part of the bottom wall **34** integrated with the protection portion **41** does not have to be shaped in such a manner and may be flat or the like.

(68) In the above embodiment, the shell portion **31** is bracket-shaped in a cross section that is perpendicular to the direction in which the shell portion **31** extends. However, the shell portion **31** only needs to include two side walls and a bottom wall that connects the ends of the two side walls. For example, the shell portion **31** may be U-shaped, arched, or annular in a cross section that is perpendicular to the direction in which the shell portion **31** extends. Alternatively, the shell portion **31** does not necessarily have to include two side walls and a bottom wall that connects the end of each of the two side walls as long as the shell portion **31** covers the outer side of the electric wires **21**. For example, the shell portion **31** may be L-shaped in a cross section that is perpendicular to the direction in which the shell portion **31** extends.

(69) The rib thickness **T1** of the first rib **45** may be different from the rib thickness **T2** of the second rib **46**. Further, the rib thickness **T1**, **T2** and the pitch **P** do not have to be set to the value indicated in the above embodiment and may be varied as required by the specification of the wire harness **20**.

(70) As viewed in the opposing direction **X2**, the cells **44** may have a hexagonal shape that is not a regular hexagon.

(71) In the above embodiment, the protection portion **41** has a honeycomb structure. However, the protection portion **41** does not necessarily have to have a honeycomb structure. For example, the cells **44** may be circular, semicircular, elliptic, or have a polygonal shape that is not a regular hexagon as viewed in the opposing direction **X2**. Further, the cells **44** may be arranged at irregular intervals.

(72) In the above embodiment, the cells **44** extend in the protection portion **41** in the opposing direction **X2**. However, the cells **44** may extend in the protection portion **41** in a different direction. For example, the cells **44** may extend in a direction that is perpendicular to both the opposing direction **X2** and the direction in which the shell portion **31** extends. Alternatively, the cells **44** may extend in the direction in which the shell portion **31** extends.

(73) In the above embodiment, the protection portion **41** disperses and absorbs the impact applied to the protection portion **41**. Alternatively, instead of positively dispersing an impact, the protection portion **41** may be increased in rigidity to withstand and dampen the impact applied from outside the protector to the electric wires **21**.

(74) In the above embodiment, the material of the protector **22** is, for example, polyamide. However, the material of the protector **22** is not limited to polyamide and may be any resin.

(75) In the above embodiment, five electric wires **21** are covered by the shell portion **31**. However, one will be sufficient as the number of the electric wires **21** covered by the shell portion **31**.

(76) In the above embodiment, the wire harness **20** is laid out under the floor or the like of the vehicle **10**. However, the wire harness **20** may be arranged at a different location in the vehicle **10** in accordance with where the wire harness **20** is connected to the electric component **11**. Further, the protector **22** may be changed in position and orientation in the vehicle **10** according to the arrangement of the wire harness **20**.

(77) In the above embodiment, the wire harness **20** connects the electric components **11**, namely, the inverter **12** and the battery **13**. However, the electric components **11** connected by the wire harness **20** are not limited and may be any electric components **11** installed in the vehicle **10**. The electric components **11** include a controller and the like. In this manner, the wire harness according to the present disclosure is not limited to a high-voltage harness and may also be a low-voltage harness.

(78) The present disclosure includes the following examples. Reference numerals of the components of the embodiment are given to facilitate understanding and not to limit the scope of the invention.

Embodiment 1

(79) A protector (22) in accordance with an example of the present disclosure may be a protector (22) that is formed from a resin and covers one or more electric wires (21), the protector (22) including: a first end and a second end in a longitudinal direction (X1) of the protector (22); an electric wire accommodation portion (35) extending from the first end to the second end and defining an open space to accommodate the one or more electric wires; a protection portion (41) arranged between the first end and the second end; and a partition wall (bottom wall 34) extending in the longitudinal direction (X1) to partition the electric wire accommodation portion (35) and the protection portion (41).

Embodiment 2

(80) In one or more examples of the present disclosure, the electric wire accommodation portion (35) may be gutter-shaped and be open in a direction opposite to the partition wall, and the protector (22) may further include a lid (51) that covers the opening.

Embodiment 3

(81) In one or more examples of the present disclosure, each of the first end and the second end may be an open end, and the protection portion 41 (41) may be separated from the first end and the second end.

Embodiment 4

(82) In the one or more examples of the present disclosure, the protection portion (41) may have a honeycomb structure.

Embodiment 5

(83) In the one or more examples of the present disclosure, the protection portion (41) may include through cells (44) extending through the protector (22) in a widthwise direction.

Embodiment 6

(84) In the one or more examples of the present disclosure, the protection portion (41) may include a flat outer wall (impact receiving wall 42) defining an outer surface of the protector (22).

Embodiment 7

(85) In the one or more examples of the present disclosure, the protector (22) may be bent at one or more locations in the longitudinal direction (X1).

Embodiment 8

(86) In the one or more examples of the present disclosure, the protection portion (41) may have a dimension that is greater than a dimension of the electric wire accommodation portion (35) in a thickness-wise direction of the protector (22).

Embodiment 9

(87) In the one or more examples of the present disclosure, the partition wall (34) may be curved in an arcuate manner in a cross section taken along a plane intersecting the longitudinal direction (X1) to increase a volume of the electric wire accommodation portion (35).

REFERENCE SIGNS LIST

(88) 10 vehicle 11 electric device 12 inverter 13 battery (electric power supply) 20 wire harness 21 electric wire 22 protector 31 shell portion 32 side wall 33 side wall 34 bottom wall 35 electric wire accommodation portion 36 protection recess 36a bottom 36b side wall 41 protection portion 42 impact receiving wall 43 impact receiving surface 44 cell 45 first rib 46 second rib 51 lid T1 rib thickness T2 rib thickness P pitch X1 shell-portion extending direction X2 opposing direction

Claims

1. A protector, comprising: a shell portion that is formed from a resin and covers an outer side of an electric wire; and a protection portion that dampens an impact applied from outside the protector, wherein the protection portion is formed from a resin and formed integrally with the shell portion, the shell portion is gutter-shaped and includes two opposing side walls in a width direction and a bottom wall connecting an end of each of the two side walls, the bottom wall has a protection recess recessed toward the inside of the shell portion, the protection portion is arranged in the protection recess and formed integrally with peripheral surfaces in the protection recess, the protection portion includes cells extending parallel to one another, and the cells are through holes extending through the protection portion in the width direction.
 2. The protector according to claim 1, wherein the protection portion disperses and absorbs the impact applied to the protection portion.
 3. The protector according to claim 1, wherein the protection portion has a honeycomb structure in which the cells are hexagonal.
 4. The protector according to claim 1, wherein the protection portion is formed integrally with a bottom of the protection recess that is part of the bottom wall, and a part of the bottom wall that is integrated with the protection portion is curved in an arcuate manner and bulged toward the protection portion in a cross section that is perpendicular to a direction in which the shell portion extends.
 5. The protector according to claim 1, wherein the protection portion is integrated with a part of an outer surface of the shell portion, and the protection portion includes a flat side surface facing a side opposite to the shell portion and exposed to the outside.
 6. A wire harness, comprising: an electric wire electrically connected to an electric power supply that drives a power source for moving a vehicle; and the protector according to claim 1, including the shell portion that covers the outer side of the electric wire.
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